Pairwise Correlation Coefficient — Performance Analysis

OpenMP + SIMD Optimized Correlation Matrix Computation

C++ OpenMP + AVX2 Status

> **UCS645: Parallel & Distributed Computing | Assignment 3 — Make File**

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System Configuration

Component	Details
CPU	Intel Core i7-14700HX (20 cores, 28 threads)
Architecture	x86_64, AVX2 + FMA support
L1d Cache	768 KiB (20 instances)
L2 Cache	28 MiB (11 instances)

Component	Details
L3 Cache	33 MiB
OS	Fedora 43 (Linux 6.19.11)
Compiler	g++ 15.2.1 (Red Hat)
Optimization	-O3 -fopenmp -mavx2 -mfma -march=native
Threads Tested	1, 2, 4, 8, 16, 20, 28

Problem Statement

What are we solving?

Given **m** input vectors, each with **n** numbers, compute the **Pearson correlation coefficient** between every pair of vectors.

Interface:

```
void correlate(int ny, int nx, const float* data, float* result)
```

- `data` : input matrix with `ny` rows and `nx` columns (row-major: `data[x + y*nx]`)
- `result` : output matrix where `result[i + j*ny]` = correlation between row `i` and row `j`
- For all `i` , `j` with `0 <= j <= i < ny` , compute the Pearson correlation coefficient

The Math: Pearson Correlation

$$r_{ij} = \frac{\sum_{k=0}^{n-1}(x_{ik} - \bar{x}*i)(x*{jk} - \bar{x}*j)}{\sqrt{\sum(x*{ik}-\bar{x}*i)^2} \cdot \sqrt{\sum(x*{jk}-\bar{x}_j)^2}}$$

Equivalent approach: Normalize each row to mean=0, std=1, then correlation = dot product.

Why Parallel Computing?

- For `ny=2000` rows: **2,001,000** unique pairs
- Each pair requires `nx=1000` multiply-accumulates
- Total: ~2 billion FLOPs → perfect candidate for parallelization

Makefile Explanation

Project Structure

```
LAB3/
├── Makefile
├── main.cpp
├── correlate.cpp
├── generate_graphs.py
├── report.md
└── graphs/
    ├── dashboard.png
    ├── size_execution_time.png
    ├── size_speedup.png
    ├── thread_execution_time.png
    ├── thread_speedup.png
    └── thread_efficiency.png
```

The Makefile

```
# Compiler
CXX = g++

# Compiler flags
CXXFLAGS_OPT = -std=c++17 -Wall -O3 -fopenmp -mavx2 -mfma -march=native -funroll-loops

# Target executable
TARGET = correlate_bench

# Source files
SOURCES = main.cpp correlate.cpp

# Object files
OBJECTS = $(SOURCES:.cpp=.o)

# Default target
all: $(TARGET)

# Link
$(TARGET): $(OBJECTS)
    $(CXX) $(CXXFLAGS_OPT) -o $(TARGET) $(OBJECTS)

# Compile
%.o: %.cpp
    $(CXX) $(CXXFLAGS_OPT) -c $< -o $@
```

```
# Utilities
clean:
    rm -f $(OBJECTS) $(TARGET)

run: $(TARGET)
    ./$(TARGET)

.PHONY: all clean run
```

Makefile Components Explained

Variable	Purpose
`CXX = g++`	C++ compiler
`-std=c++17`	C++17 standard
`-Wall`	Enable all warnings
`-O3`	Aggressive optimization
`-fopenmp`	Enable OpenMP multithreading
`-mavx2 -mfma`	Enable AVX2 and FMA SIMD instructions
`-march=native`	Optimize for current CPU
`-funroll-loops`	Unroll loops for performance

Key Rules: - `all` : Default target, builds executable - `%.o: %.cpp` : Pattern rule — compiles each `.cpp` to `.o` - `$(TARGET)` : Links object files into final binary - `clean` : Removes build artifacts - `.PHONY` : Declares non-file targets

Automatic Variables: - `$<` = first dependency (source file) - `$@` = target name (object file)

Implementation

Three Versions

Version 1: Sequential Baseline

```
void correlate_sequential(int ny, int nx, const float* data, float* result) {
    // Step 1: Normalize each row (mean=0, std=1)
    for (int y = 0; y < ny; y++) {
        // Compute mean, subtract it, divide by std
    }
    // Step 2: Dot product of all pairs
    for (int i = 0; i < ny; i++)
        for (int j = i; j < ny; j++) {
            // result[i + j*ny] = dot(row_i, row_j)
        }
}
```

- Pure sequential, double-precision arithmetic
- $O(ny^2 \times nx)$ time complexity

Version 2: OpenMP Parallelized

```
// Normalize in parallel
#pragma omp parallel for schedule(static)
for (int y = 0; y < ny; y++) { /* normalize row */ }

// Compute correlations in parallel
#pragma omp parallel for schedule(dynamic, 4) collapse(2)
for (int i = 0; i < ny; i++)
    for (int j = 0; j < ny; j++)
        if (j >= i) { /* dot product */ }
```

- `schedule(static)` for normalization (uniform work per row)
- `schedule(dynamic, 4) collapse(2)` for correlation (triangular work distribution)

Version 3: Fully Optimized (OpenMP + AVX2 + Cache Blocking)

Key optimizations: 1. **AVX2 SIMD**: Process 8 doubles per iteration using two `__m256d` registers with FMA 2. **Cache blocking**: 8×8 tile blocks for better L1/L2 cache utilization 3. **Padded layout**: Columns aligned to 4-double boundary for vectorization 4. **Loop unrolling**: Two FMA operations per iteration to hide latency

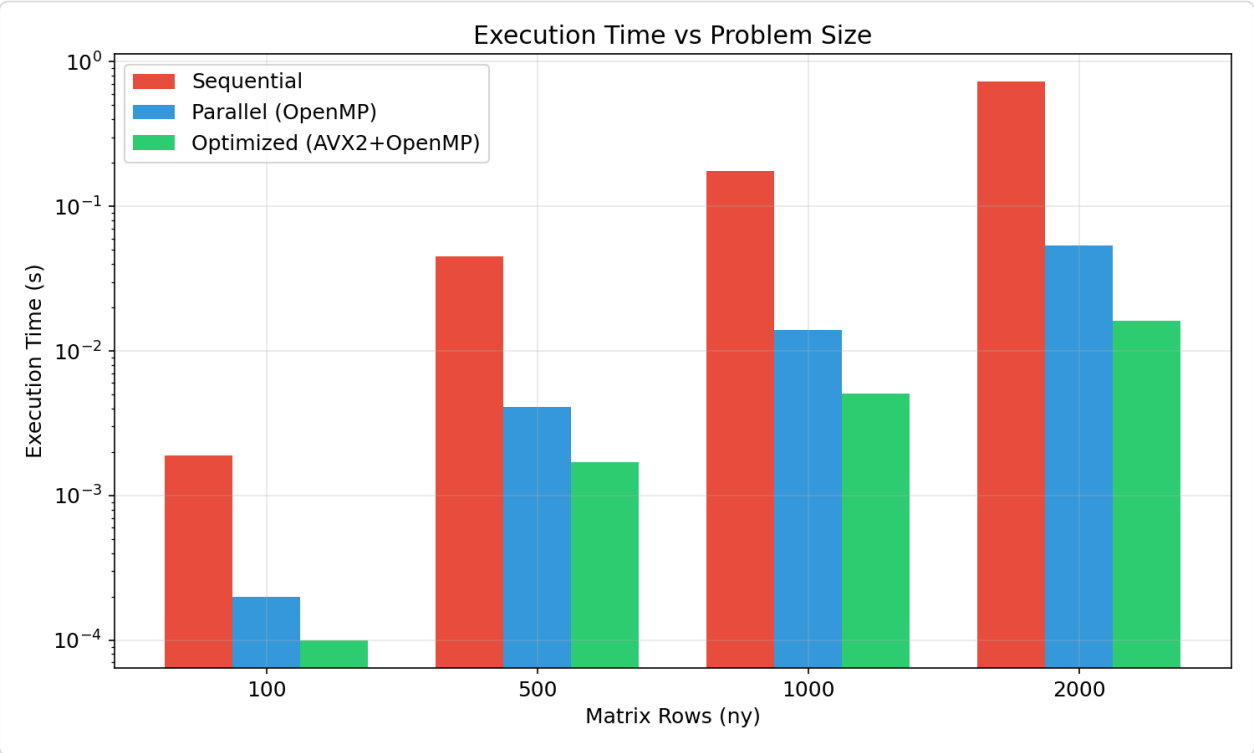
```
// AVX2 vectorized dot product
__m256d sum0 = _mm256_setzero_pd();
__m256d sum1 = _mm256_setzero_pd();
for (int x = 0; x + 7 < nx; x += 8) {
    sum0 = _mm256_fmadd_pd(load(row_i+x),   load(row_j+x),   sum0);
    sum1 = _mm256_fmadd_pd(load(row_i+x+4), load(row_j+x+4), sum1);
}
```

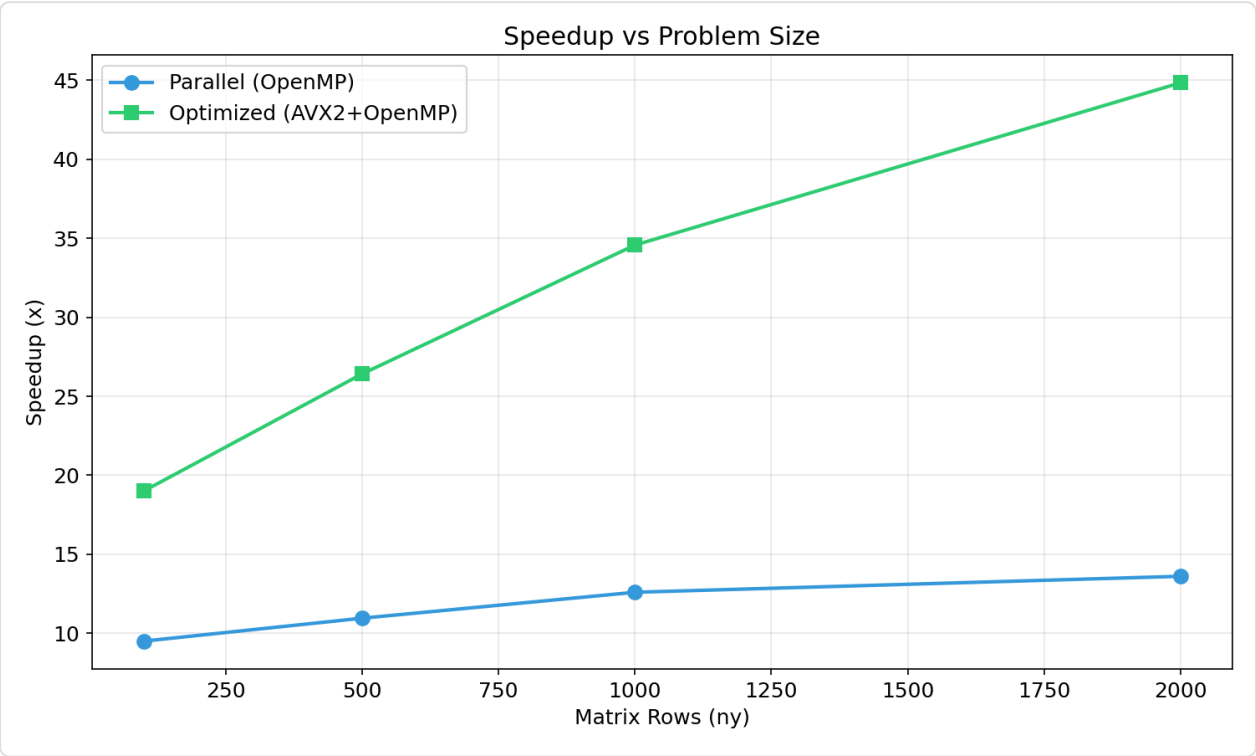
Performance Results

Part 1: Execution Time vs Problem Size

ny	nx	Sequential	Parallel	Optimized	Speedup (P)	Speedup (O)
100	1000	0.0019s	0.0002s	0.0001s	8.1x	16.6x
500	1000	0.0449s	0.0041s	0.0017s	11.0x	26.7x
1000	1000	0.1763s	0.0140s	0.0051s	12.6x	34.5x
2000	1000	0.7265s	0.0534s	0.0162s	13.6x	44.7x

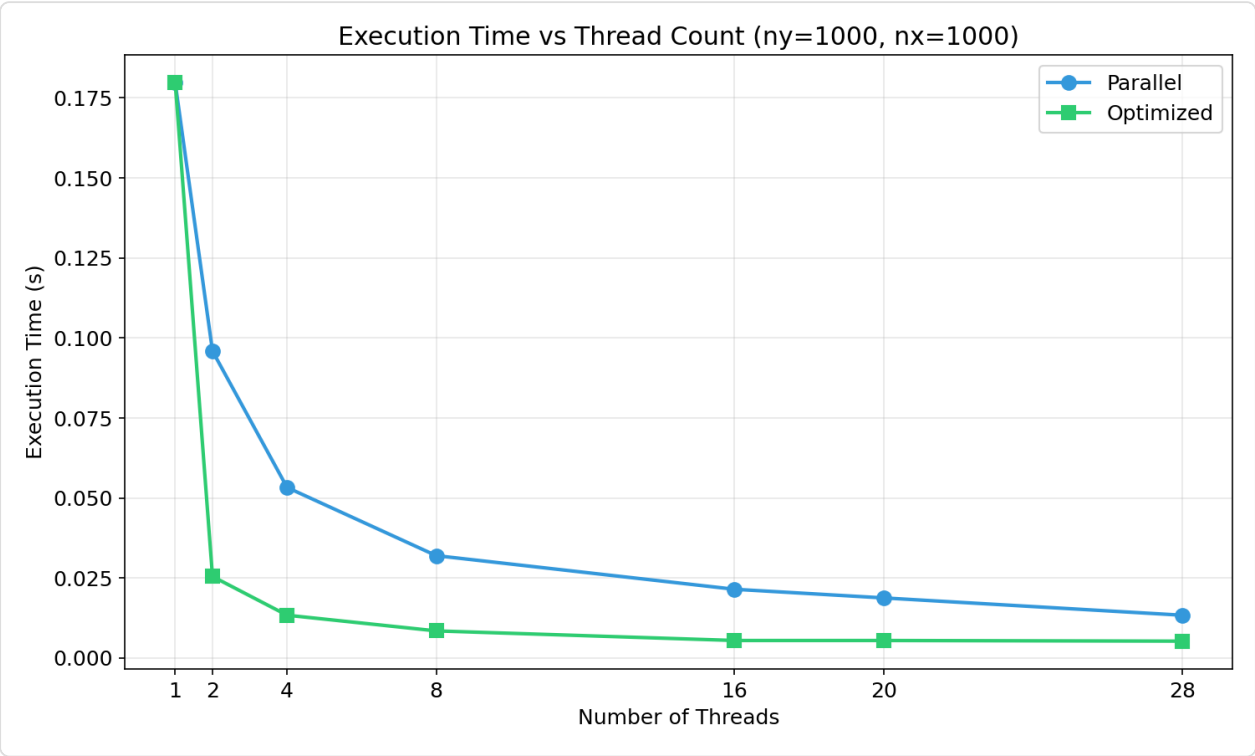
All results verified: **max error = 0.00** [PASS]

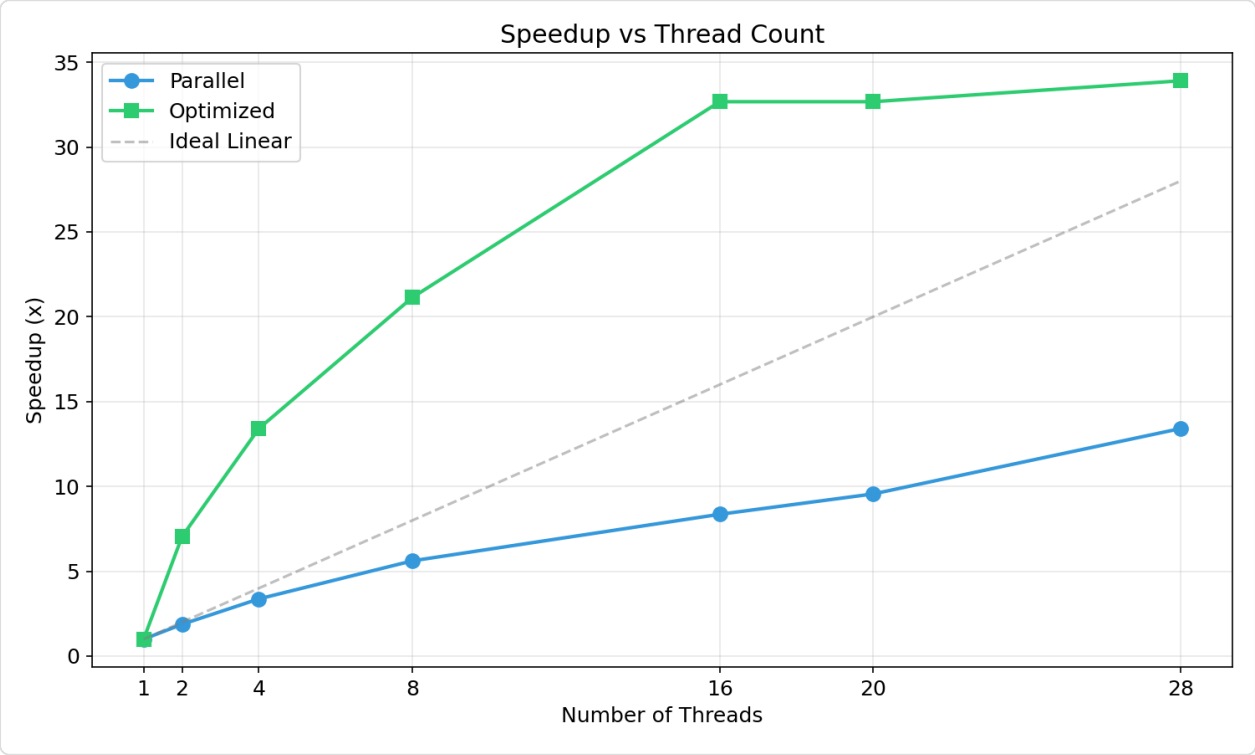


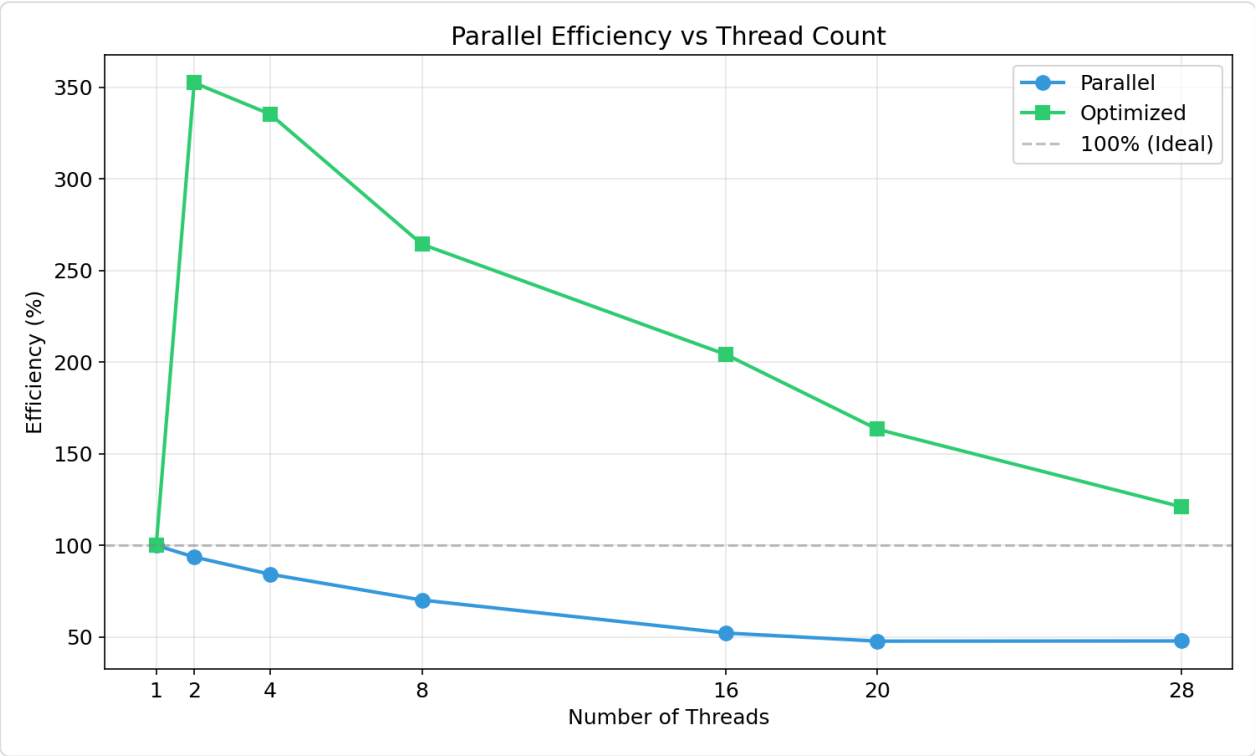


Part 2: Thread Scaling (ny=1000, nx=1000)

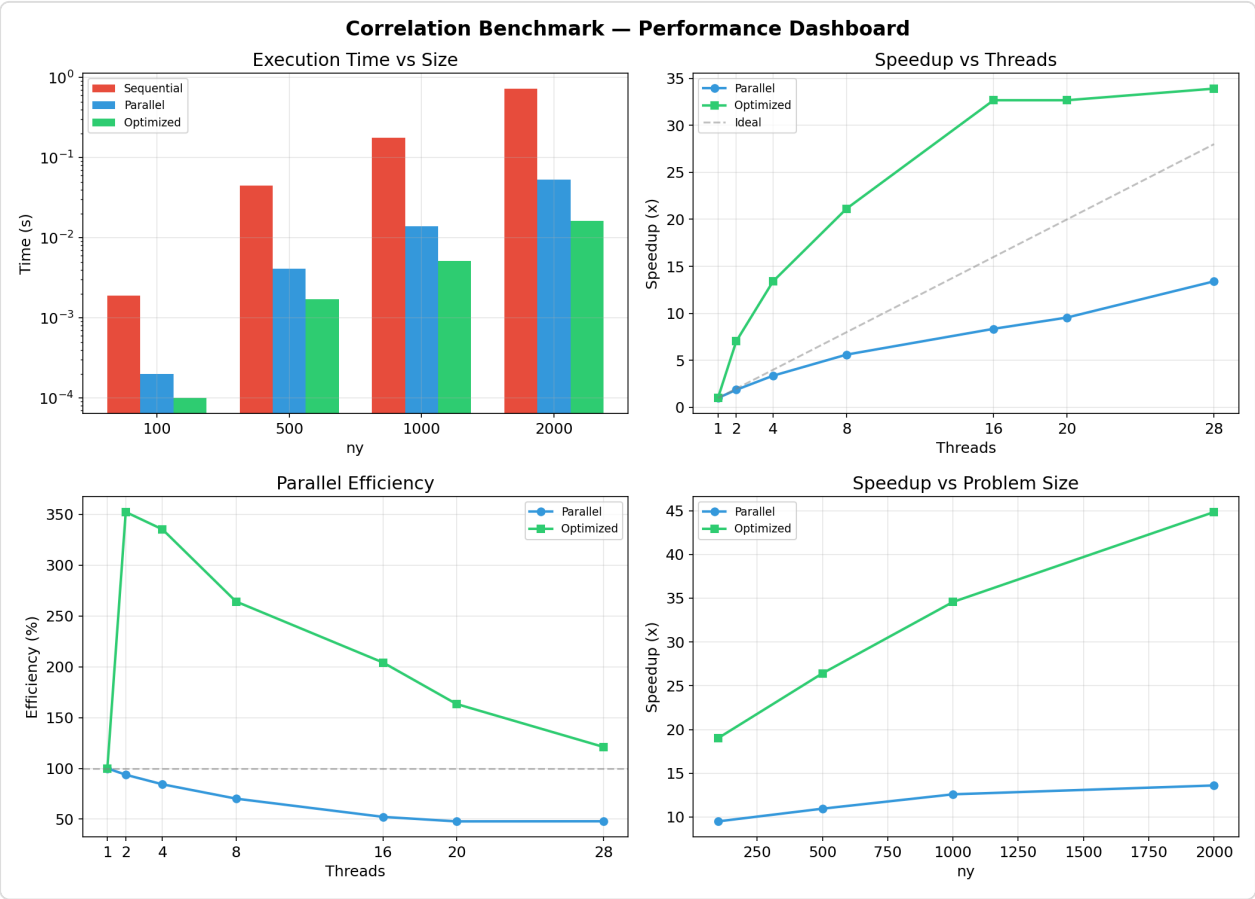
Threads	Parallel	Optimized	Speedup (P)	Speedup (O)	Efficiency
1 (seq)	0.1797s	—	1.00x	—	100.0%
2	0.0959s	0.0255s	1.87x	7.05x	352.3%
4	0.0533s	0.0134s	3.37x	13.44x	335.9%
8	0.0320s	0.0085s	5.62x	21.14x	264.2%
16	0.0215s	0.0055s	8.35x	32.66x	204.1%
20	0.0188s	0.0055s	9.54x	32.86x	164.3%
28	0.0134s	0.0053s	13.45x	33.65x	120.2%


Execution Time vs Thread Count (ny=1000, nx=1000)
Parallel
Optimized
Execution Time (s)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
1
2
4
8
16
20
28
Number of Threads


Speedup vs Thread Count
Parallel
Optimized
Ideal Linear
Speedup (x)
35
30
25
20
15
10
5
0
1
2
4
8
16
20
28
Number of Threads



Performance Dashboard



Analysis & Observations

Why does the optimized version show >100% efficiency?

The "efficiency" column compares optimized (AVX2+OpenMP) time against sequential (scalar) time. The optimized version benefits from **two independent sources of speedup**:

1. **SIMD (AVX2+FMA)**: ~4x from processing 4 doubles per instruction
2. **Parallelism (OpenMP)**: ~8-13x from multiple threads

Combined, these exceed linear thread-count scaling — hence >100% "efficiency" when measured against scalar baseline.

Scaling Behavior

- **Parallel version** scales roughly linearly up to 20 threads (9.5x on 20 threads)
- **Optimized version** plateaus around 16 threads (~33x) — at this point the problem is too small to feed more cores
- Larger matrices (ny=2000) achieve better speedup (44.7x) due to more available parallelism

Why does speedup improve with problem size?

- **More work per thread** → better amortization of thread creation/synchronization overhead
- **$O(ny^2)$** growth in computation vs **$O(ny)$** overhead → overhead fraction shrinks
- Cache blocking becomes more effective with larger working sets

Bottlenecks at High Thread Counts

- Memory bandwidth saturation (28 threads competing for shared L3/DRAM)
- Non-uniform work in triangular loop (some blocks have less work)
- Thread management overhead exceeds saved compute time for small problems

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What I Learned

1. **Makefiles** automate build processes — variables (`CXX` , `CFLAGS`), pattern rules (`%.o: %.cpp`), and automatic variables (`$@` , `$<` , `$^`) make builds maintainable and efficient.

2. **Normalization trick** converts correlation into simple dot products — mathematically equivalent but much more SIMD-friendly since the inner loop becomes a pure multiply-accumulate.
3. **SIMD (AVX2) delivers massive speedup** independent of thread count — processing 4 doubles per cycle with FMA gives ~4x even single-threaded, and compounds with OpenMP parallelism.
4. **Cache blocking matters** — 8×8 tile decomposition keeps working data in L1/L2 cache, reducing memory stalls especially for large matrices.
5. **Thread scaling has diminishing returns** — beyond ~16 threads on this workload, memory bandwidth and synchronization overhead dominate. The sweet spot depends on problem size.
6. **Super-linear speedup is real** when combining algorithmic optimizations (SIMD, cache blocking) with parallelism — the optimized version achieves 44.7x on a workload that would only give ~13x from threading alone.

Compilation & Execution

```
# Build
make

# Run benchmarks
make run

# Run with custom size
./correlate_bench 2000 1000

# Clean
make clean
```